SOC Lab Report: Privilege Escalation Detection using Splunk

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1 Abstract

This report demonstrates detection of privilege escalation in a SOC environment using Splunk. Privilege escalation occurs when a user gains elevated permissions such as administrative access. This scenario simulates adding a user to the administrators group and detecting it using Windows event logs.

2 Objective

To detect unauthorized privilege escalation by monitoring administrative group changes using Splunk.

3 Lab Setup

- Attacker Machine: Kali Linux
- Victim Machine: Windows 11
- SIEM Tool: Splunk (Ubuntu Server)
- Network: Internal Virtual Lab

4 MITRE ATT&CK Mapping

- Tactic: Privilege Escalation
- Technique: Account Manipulation (T1098)

5 Attack Simulation

Privilege escalation was simulated by adding a user to the administrators group.

5.1 Commands Used

```
net user testuser Password123 /add
net localgroup administrators testuser /add
```

5.2 Explanation

A new user account was created and added to the administrators group, granting elevated privileges.

5.3 Screenshot

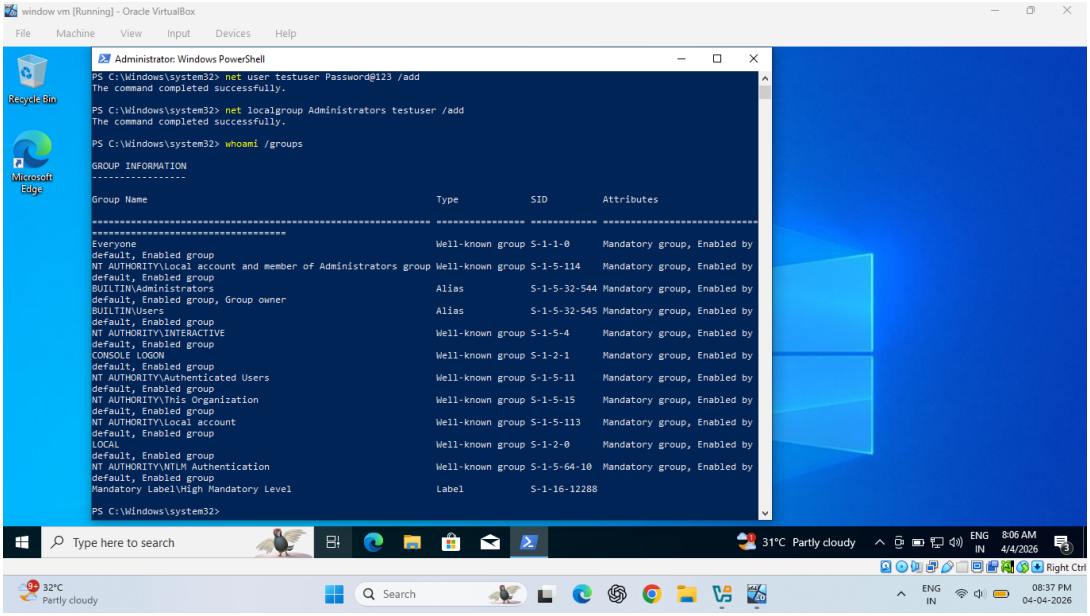

Figure 1: User Added to Administrators Group

6 Detection in Splunk

Privilege escalation is detected using Windows Security Event Codes.

6.1 Detection Query

```
index=windows EventCode=4728 OR EventCode=4672
| table _time, user, host
```

6.2 Explanation

EventCode 4728 indicates a user added to a privileged group, and EventCode 4672 indicates special privileges assigned.

6.3 Screenshot

7 Investigation

Further investigation was conducted to identify unauthorized privilege escalation.

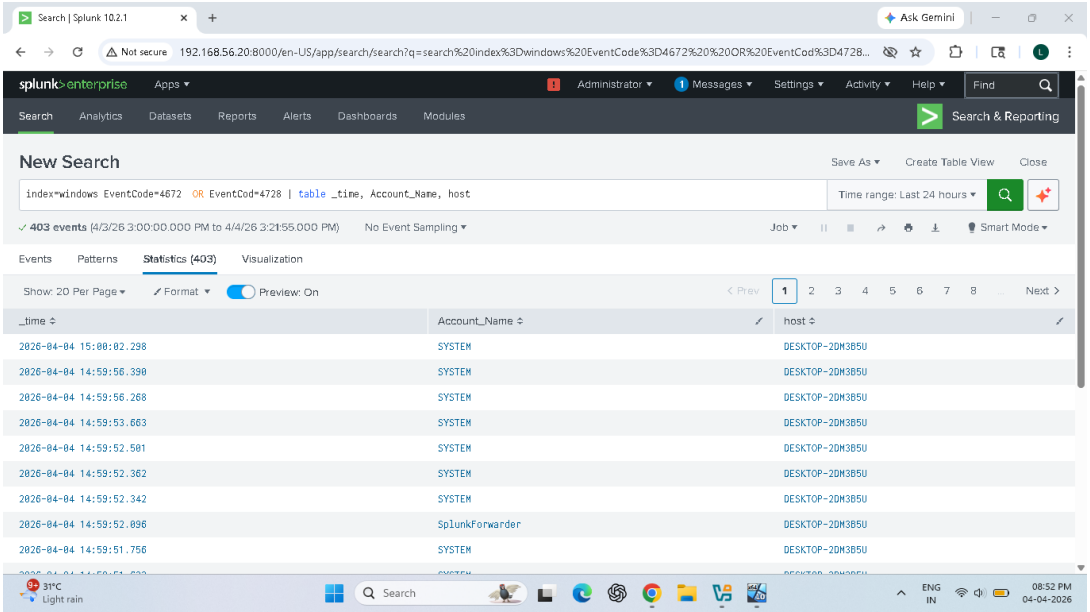

Figure 2: Detection of Privilege Escalation in Splunk

7.1 Timeline Query

```
index=windows user="testuser"
| sort _time
| table _time, EventCode, Command_Line
```

7.2 Findings

- User `testuser` was added to administrators group
- Privileged access granted without authorization
- Potential attacker activity detected

7.3 Screenshot

7.4 Screenshot

7.5 Screenshot

8 Response Actions

Immediate actions were taken to mitigate the risk.

- Removed user from administrators group
- Disabled the user account
- Reset credentials

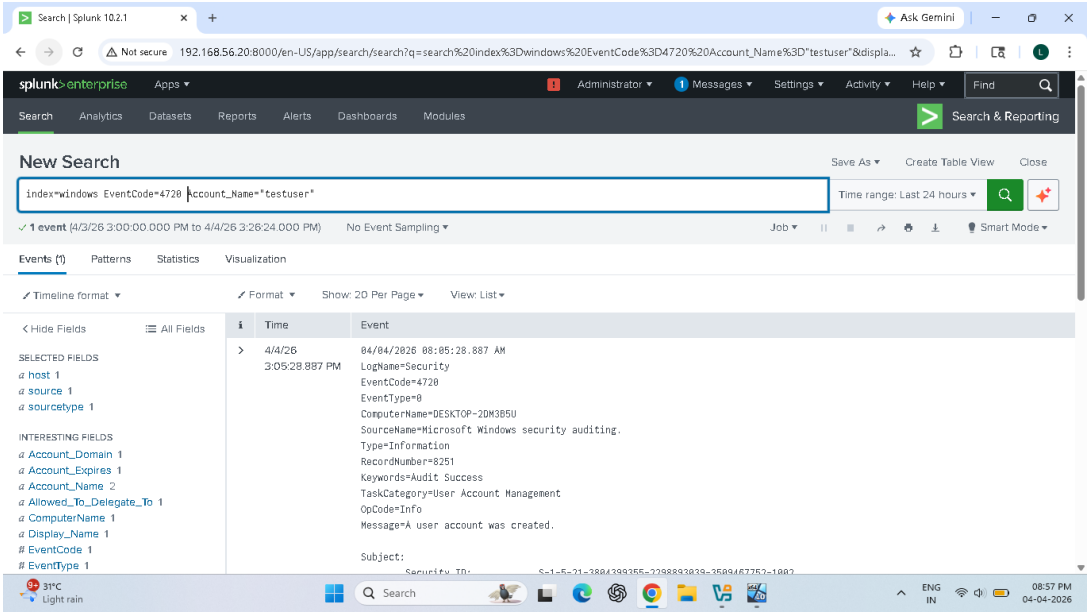

Figure 3: Timeline showing privilege escalation

8.1 Commands Used

```
net localgroup administrators testuser /delete
net user testuser /active:no
net user testuser NewPassword@123
```

8.2 Explanation

The elevated privileges were revoked, and the user account was secured to prevent further misuse.

8.3 Screenshot

9 Incident Reporting

9.1 Incident Summary

A user account was granted administrative privileges, indicating a privilege escalation attempt.

9.2 Impact

Unauthorized administrative access could lead to full system compromise.

9.3 Conclusion

The escalation was detected and mitigated using Splunk and appropriate response actions.

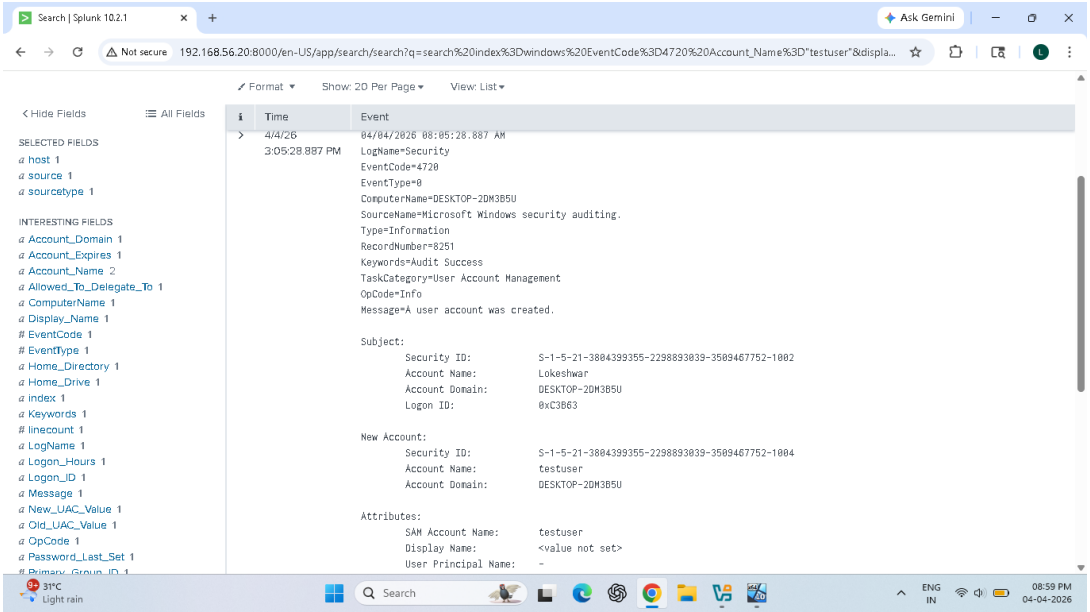

Figure 4: Timeline showing privilege escalation

10 Conclusion

This scenario highlights the importance of monitoring privileged group changes to prevent unauthorized access.

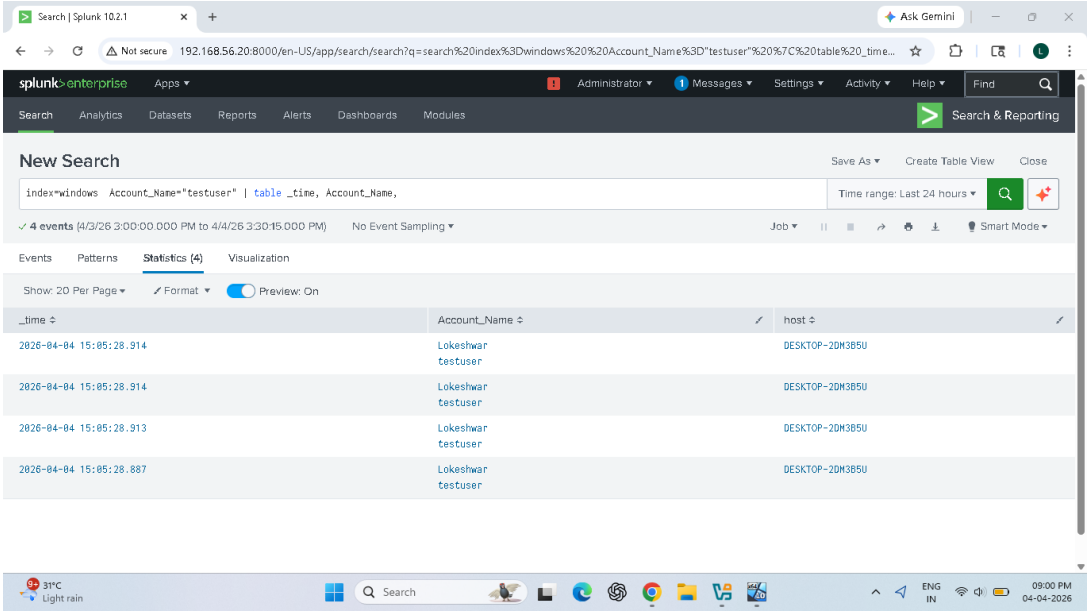

Figure 5: Timeline showing privilege escalation

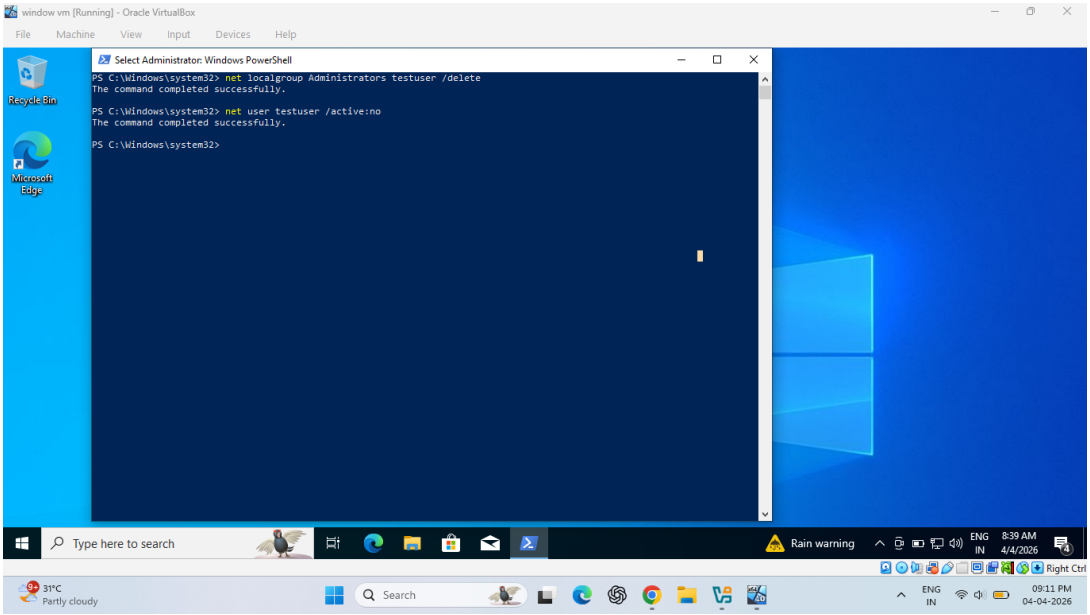

Figure 6: Response Actions - Removing Admin Access